

Empirical Study

3. How do policies conditioned on automation and/or instruction embeddings perform and generalize?

3. RAD Embeddings enable optimal performance and generalization across task classes.



Train data; Emb type	ReachAvoid (Train)	Reach (Test)	ReachAvoid (Test)	RAD (Test)
RA-400; RAD Emb	1.0 $\pm$ 0.0	0.94 $\pm$ 0.02	0.97 $\pm$ 0.01	0.98 $\pm$ 0.01
RA-400; Qwen3 Emb	0.72 $\pm$ 0.06	0.77 $\pm$ 0.05	0.60 $\pm$ 0.06	0.74 $\pm$ 0.06
RA-400; Both	1.0 $\pm$ 0.0	0.94 $\pm$ 0.02	0.98 $\pm$ 0.01	0.98 $\pm$ 0.01
RA-100; RAD Emb	1.0 $\pm$ 0.0	0.94 $\pm$ 0.03	0.93 $\pm$ 0.01	0.96 $\pm$ 0.02
RA-100; Qwen3 Emb	0.96 $\pm$ 0.01	0.59 $\pm$ 0.06	0.55 $\pm$ 0.03	0.63 $\pm$ 0.03
RA-100; Both	1.0 $\pm$ 0.0	0.94 $\pm$ 0.03	0.92 $\pm$ 0.03	0.95 $\pm$ 0.02

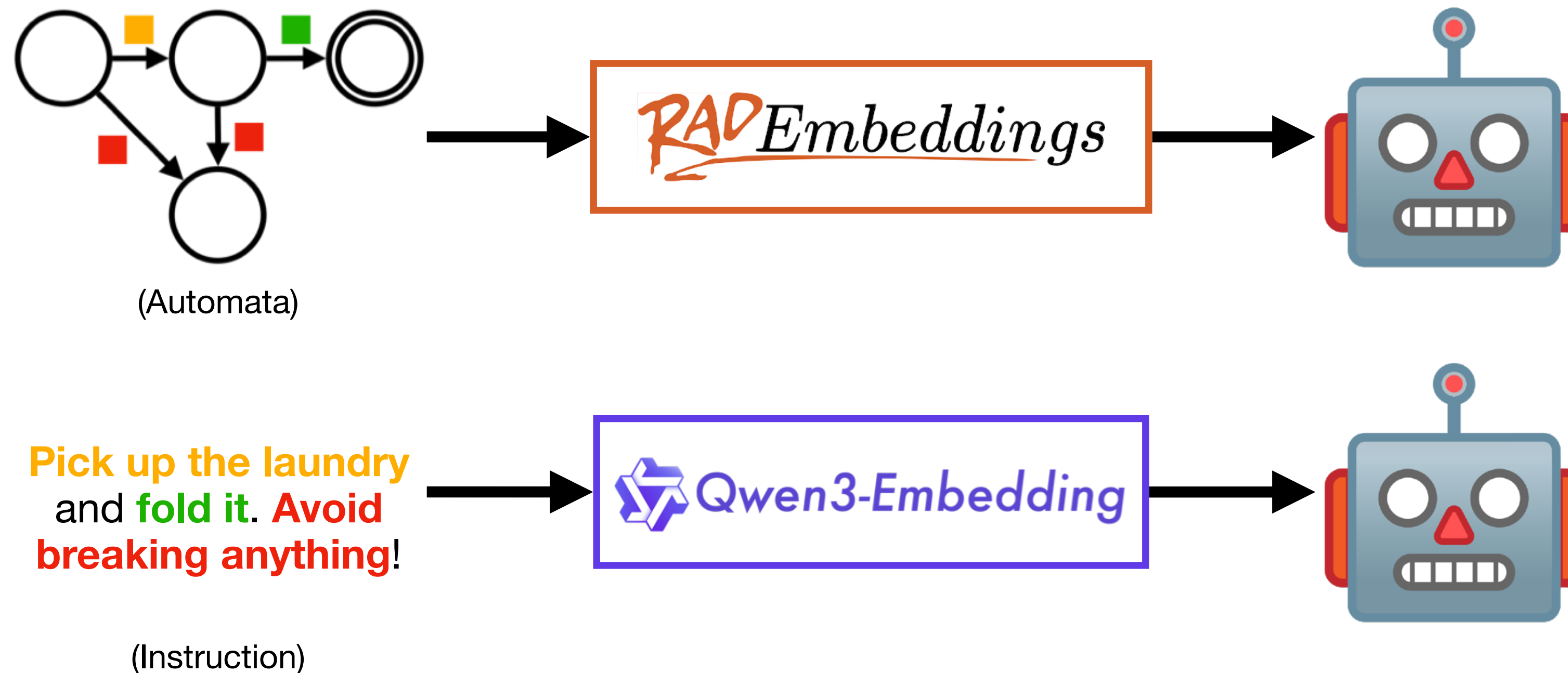
Empirical Study

3. How do policies conditioned on automaton and/or instruction embeddings perform and generalize?

Train data; Emb type	ReachAvoid (Train)	Reach (Test)	ReachAvoid (Test)	RAD (Test)
RA-400; RAD Emb	1.0 ± 0.0	0.94 ± 0.02	0.97 ± 0.01	0.98 ± 0.01
RA-400; Qwen3 Emb	0.72 ± 0.06	0.77 ± 0.05	0.60 ± 0.06	0.74 ± 0.06
RA-400; Both	1.0 ± 0.0	0.94 ± 0.02	0.98 ± 0.01	0.98 ± 0.01
RA-100; RAD Emb	1.0 ± 0.0	0.94 ± 0.03	0.93 ± 0.01	0.96 ± 0.02
RA-100; Qwen3 Emb	0.96 ± 0.01	0.59 ± 0.06	0.55 ± 0.03	0.63 ± 0.03
RA-100; Both	1.0 ± 0.0	0.94 ± 0.03	0.92 ± 0.03	0.95 ± 0.02

3. RAD Embeddings enable optimal performance and generalization across task classes.

Recap: Our goal was to compare these two modalities.



1. Which modality is better for learning-based control?
 2. What happens when these modalities are combined?
- **Challenge:** How to generate automaton-instruction pairs?